Ovarian Mass Assessment App: Build Spec for Claude Code

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Purpose and scope

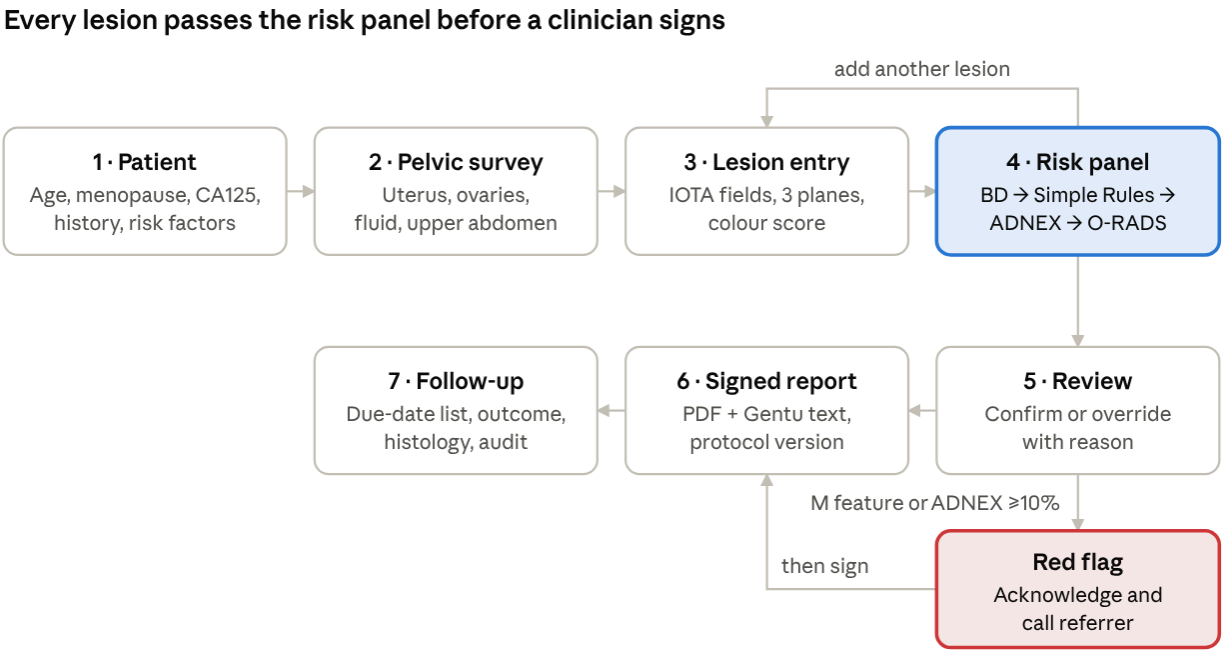
A secure web app that takes a sonographer through the CFWU ovarian mass protocol step by step and records IOTA findings. It runs the two-step strategy, assigns an O-RADS US v2022 category, and produces a signed structured report with the protocol's management recommendation.

The clinical rules come from Ovarian Mass Ultrasound: Guideline Review & Scanning and Reporting Protocol. The app implements that protocol and changes only when the protocol changes.

- **Users:** sonographers and WSU student sonographers (data entry), sonologists and MFM/O&G specialists (review and sign), practice admin (audit exports).
- **Sites:** Westmead Innovation Quarter and Norwest Private Hospital, Bella Vista.
- **In scope (v1):** patient and clinical context, per-lesion IOTA description, modified benign descriptors, Simple Rules, ADNEX risk (via a validated implementation — see Clinical logic), O-RADS category, management text, structured report (PDF and copy-to-clipboard), follow-up tracking and audit dashboard.
- **Out of scope (v1):** image analysis or automatic measurement from ultrasound images (keeps the app outside TGA image-processing rules), direct writes into Gentu or Viewpoint (phase 3), and patient-facing access.
- **Design principle:** the app *suggests*; the reporting clinician always confirms or overrides the category and recommendation, with the reason recorded.

Clinical workflow

The app mirrors the protocol's scanning order: context, survey, one form per lesion, a live risk panel, then clinician review before the report can be signed.



App workflow · 7 screens, per-lesion loop, red-flag gate

The risk panel updates as fields are entered, so the sonographer sees the BD, Simple Rules and O-RADS result before the patient leaves; a red flag blocks signing until it is acknowledged and the referrer call is logged.

Clinical logic

All rules live in one versioned, clinician-approved rules file (`rules/protocol-v1.json`) with unit tests. They are never hard-coded across screens, and Claude Code must not invent any coefficient, threshold or management text.

Rule engine, in order (per lesion)

1. **Physiological check:** clinician ticks "follicle / corpus luteum ⩽3 cm, premenopausal" → O-RADS 1; stop.
2. **Modified benign descriptors** (deterministic; all need largest diameter <10 cm):
   - BD1 = unilocular AND ground-glass AND premenopausal
   - BD2 = unilocular AND mixed echogenicity AND acoustic shadows AND premenopausal
   - BD3 = unilocular AND anechoic AND regular walls
   - BD4 = unilocular AND regular walls (any other content)

   Any true → O-RADS 2, show which BD applied.

3. **ADNEX** (when no BD applies): inputs are age, CA125 (optional; run the with/without model accordingly), centre type, maximum lesion diameter, maximum solid component diameter, >10 locules, papillary projection count (0/1/2/3/>3), acoustic shadows, ascites. Outputs are total risk plus benign, borderline, stage I, stage II–IV and metastatic probabilities. If the solid component is larger than the lesion, block calculation and ask for correction.
4. **Simple Rules** (deterministic): B1–B5 and M1–M5 from the same fields → benign / malignant / inconclusive. Always shown; any M feature raises an escalation flag.
5. **RMI U-score** features listed for referrer compatibility; RMI calculated only if CA125 is entered.
6. **O-RADS category:** from the ADNEX band using the IOTA 2026 mapping (<1% = 2; 1–<10% = 3; 10–<50% = 4; ≥50% = 5). The clinician can also assign by O-RADS descriptors; the higher category displays with both shown.
7. **Escalation flags** (red banner, report cannot be finalised without acknowledgement): any M feature; ADNEX ≥10%; ADNEX metastatic risk ≥10%; ascites; RMI >200.
8. **Management text:** pulled from the protocol's management table for the final category and menopausal status. The clinician can edit it, and any edit is logged as an override.

What must be sourced and verified before clinical use

Item	Source	Verification
ADNEX model coefficients (both versions)	Van Calster et al., BMJ 2014;349:g5920 and supplementary material, or directly from the IOTA group	Match the official IOTA app/online calculator to 0.1% on at least 50 test cases spanning all categories
Simple Rules and BD definitions	IOTA 2026 updated terms (UOG 2026;67:702–724); Landolfo 2023	Clinician sign-off on the rules file
O-RADS v2022 classic-lesion and follow-up tables	ACR O-RADS US v2022 assessment categories PDF	Transcribed by a clinician; second clinician checks
Management wording	CFWU protocol (reporting section)	Protocol owner approval; version number printed on every report

Fallback if ADNEX coefficients cannot be confirmed: v1 records an ADNEX result entered manually from the IOTA app (number + subclass percentages) instead of calculating it. Everything else still runs.

Data model and report output

One examination holds one or more lesions; each lesion stores its raw IOTA fields, every computed result, and the rules-file version used, so any report can be reproduced exactly later.

Entity	Key fields
Patient	internal ID, MRN (Gentu), name, DOB (age derived at exam date)
Examination	site, date/time, sonographer, reporting clinician, menopausal status, LMP/cycle day, hormones, pregnancy, symptoms, family history/BRCA/Lynch, CA125 (value, date), technique (TV/TA/TR), image quality, uterus/endometrium, free fluid (POD depth / ascites), upper-abdomen findings, status (draft / for review / signed / amended)
Lesion	side, origin, 3 diameters, IOTA type, locule count, >10 locules, wall regularity, content echogenicity, solid component present + 3 diameters, papillary count + flow, solid surface, acoustic shadows, colour score, tenderness/mobility, suggested diagnosis + confidence
Assessment (per lesion)	BD result, ADNEX total + 5 subclasses (calculated or entered), Simple Rules B/M features + result, RMI features/score, O-RADS (model-derived, descriptor-derived, final), flags, management text, override reason, rules version
Follow-up	due date, reason, linked prior lesion, outcome (resolved / stable / surgery), histology result
Audit log	who changed what and when, including every override and amendment

Report output: generated from the protocol's structured template — clinical details, per-lesion IOTA description, risk assessment block (BD, ADNEX with subclasses, Simple Rules, RMI features, O-RADS), conclusion with the recommendation and timeframe, interval

comparison, reporting clinician, sonographer, protocol version, and time of verbal communication for O-RADS 4–5. Delivered as a PDF, as plain text to paste into the Gentu report, and (phase 3) as an HL7/FHIR message.

Audit dashboard: completeness of IOTA fields, distribution by O-RADS category, override rate, time to referrer call for O-RADS 4–5, and malignancy rate per category vs histology/follow-up — the annual audit measures in the protocol.

Regulatory, privacy and IOTA licensing

Software that calculates a cancer risk for an individual patient is likely to be a medical device. It is designed to fit the TGA's clinical decision support (CDSS) exemption, but get regulatory advice before go-live.

TGA. CDSS that meets the medical-device definition must be on the ARTG unless it meets all three exemption criteria in Schedule 4, Part 2 of the Medical Devices Regulations 2002 (TGA guidance). In summary, the software must not directly process a medical image or a signal from another device, it must only support a health professional's decision, and it must not replace their clinical judgement (HWL Ebsworth summary). An exempt CDSS must still be notified to the TGA within 30 working days of supply (TGA detailed guidance). Design choices that support the exemption:

- Clinician enters measurements manually; no image import or automatic measurement.
- Every output shows its basis (which BD, which M features, the ADNEX inputs), so the clinician can review the reasoning.
- Category and recommendation are suggestions the clinician must confirm or override.
- Clinician-only access; no patient-facing output.
- *Open question:* whether internal use within CFWU counts as "supply", and whether sharing the app with other practices or WSU changes that — confirm with a regulatory adviser.

Privacy and security. Health information is covered by the Privacy Act 1988 (APPs) and the NSW Health Records and Information Privacy Act 2002.

- Host in an Australian region; encrypt data in transit and at rest.
- Single sign-on or MFA; role-based access (sonographer, student, reporting clinician, admin).
- Full audit trail; retention aligned with CFWU's medical-records policy.
- Students see only cases they are assigned; option for de-identified teaching mode.
- Never put patient identifiers in URLs; daily backups and a tested restore.

IOTA. The IOTA terms, Simple Rules and ADNEX are published, but IOTA's conditions for embedding ADNEX in third-party software, and any trademark use of the "IOTA" name, could not be confirmed online. Contact the IOTA group before building in the calculation; until then, use the manual-entry fallback described above.

Tech stack, integrations and build plan

Build the clinical rules engine first as tested, UI-free code, then wrap it in a web app. Pilot at one site with fake data before any real patient is entered.

Suggested stack (Claude Code can propose alternatives, but keep the rules engine separate):

Layer	Choice	Why
Rules engine	TypeScript pure functions + `rules/protocol-v1.json`, Vitest unit tests	Deterministic, testable, versioned
Web app	Next.js (React, TypeScript), tablet-friendly forms	Works on scanning-room PCs and tablets
Database	PostgreSQL in an Australian region (e.g. AWS Sydney or Supabase Sydney) via Prisma	Relational audit trail; data stays onshore
Auth	Microsoft Entra ID / Google Workspace SSO with MFA, role-based access	Matches practice accounts
Reports	Server-side PDF (e.g. react-pdf) + plain-text export	Paste into Gentu now; message later
Testing	Vitest (rules), Playwright (end-to-end)	Regression safety when the protocol changes

Integrations (phase 3): pull patient demographics and referral details from Gentu, and push the finalised report back, if Gentu exposes an API or HL7/FHIR interface. Link to images in Viewpoint via accession number. Both depend on the vendor interfaces, which are still being worked through with Gentu.

Build phases:

1. **Phase 0 — Rules engine (1–2 weeks):** BD, Simple Rules, O-RADS mapping, flags and management lookup with unit tests built from protocol examples; ADNEX as a manual-entry field.

2. **Phase 1 — Clickable prototype (2–3 weeks):** all forms, live risk panel, report preview; runs locally with synthetic patients only. Clinician usability review.
3. **Phase 2 — Secure pilot (4–6 weeks):** hosting, SSO, audit log, PDF reports, follow-up list; pilot at one site in parallel with normal reporting, compare outputs.
4. **Phase 3 — Integrations:** Gentu and Viewpoint links; audit dashboard.
5. **Phase 4 — ADNEX calculation:** only after coefficients are confirmed (IOTA/BMJ 2014) and validated against the IOTA app, and IOTA permission is in place.

Timeframes are rough estimates for one developer working with Claude Code; treat them as a starting point.

Acceptance tests and Claude Code kickoff prompt

Phase 0 is done when every test below passes and a clinician has signed off the rules file; paste the prompt below into Claude Code in an empty project folder to start.

Rule-engine acceptance tests (minimum)

#	Input	Expected
1	34 y, premenopausal, unilocular, ground-glass, 45 mm	BD1 → O-RADS 2; no flags
2	30 y, premenopausal, unilocular, mixed echogenicity, shadows, 60 mm	BD2 → O-RADS 2
3	62 y, postmenopausal, unilocular, anechoic, regular walls, 40 mm	BD3 → O-RADS 2
4	45 y, unilocular anechoic, regular walls, 120 mm	No BD (≥10 cm) → ADNEX required; Simple Rules B1 benign
5	Multilocular-solid, irregular, 130 mm, ascites, colour score 4	Simple Rules malignant (M2, M4, M5); red escalation flag; cannot finalise without acknowledgement
6	Unilocular-solid, 4 papillary projections	M3 present; escalation flag
7	Solid component diameter > lesion diameter	Validation error; no result
8	Both B and M features present	Simple Rules inconclusive
9	ADNEX entered as 12% total, metastatic 11%	O-RADS 4; flags for ADNEX ≥10% and metastatic ≥10%

#	Input	Expected
10	Clinician changes O-RADS 3 → 4	Override reason mandatory; audit log entry; report shows both
11	Postmenopausal ovaries not visualised	Report states "not visualised", not "normal"
12	Any signed report	Shows protocol and rules version; amendments create a new version

Kickoff prompt for Claude Code

```
You are helping me build an internal clinical web app for the Centre for
Women's Ultrasound (CFWU) to assess and record adnexal (ovarian) masses using
IOTA terminology, the IOTA two-step strategy and O-RADS US v2022, following
our protocol. I will save the protocol and this spec as docs/protocol.md and
docs/spec.md.

Non-negotiable rules:
1. Do NOT invent any clinical coefficient, threshold, definition or
management text. All clinical rules come from docs/protocol.md and
docs/spec.md and live in rules/protocol-v1.json. If something is missing or
ambiguous, stop and ask me.
2. ADNEX is a manual-entry field in this phase (total risk + 5 subclass
probabilities). Do not implement the ADNEX formula.
3. The app suggests; the clinician confirms or overrides with a recorded
reason.
4. No real patient data. Use synthetic data only. No patient identifiers in
URLs or logs.

Phase 0 task (do this first, nothing else):
- Set up a TypeScript project with Vitest.
- Implement a pure-function rules engine: physiological check, modified
benign descriptors BD1-4, IOTA Simple Rules B1-B5 and M1-M5, RMI U-score
features (and RMI if CA125 is given), O-RADS mapping from ADNEX bands (<1% =
2, 1-<10% = 3, 10-<50% = 4, >=50% = 5), escalation flags, and management text
lookup from the rules file.
- Implement input validation (e.g. solid component cannot exceed lesion
diameter).
- Write unit tests covering the 12 acceptance tests in docs/spec.md.
- Write a short README explaining how to run the tests and where each rule
comes from in the protocol.

Before writing code, give me a plan and list any questions about the
protocol. Then build Phase 0 only.
```

How to use it: export both docs as Markdown into a `docs/` folder, open Claude Code in that folder, paste the prompt, and review its plan before it writes code. After Phase 0, ask it for Phase 1 using the build plan above.